YOLO Model Architecture, Accuracy Benchmarks & Extensibility Guide

Academic Reference Document for EviGuard AI Proctoring System

1. Comparative Accuracy & Speed Benchmarks (YOLO vs Other Models)

In real-time edge applications such as online exam proctoring, a vision model must achieve high mean Average Precision (mAP) while maintaining a high frame rate (>25 FPS) on commodity hardware without excessive GPU reliance. Below is a comprehensive comparative evaluation against alternative computer vision architectures:

Model Architecture	Type / Paradigm	COCO mAP (50-95)	Inference (CPU / GPU)	AI Proctoring Viability
YOLOv8 Nano (Ultralytics)	Single-Stage (Anchor-Free)	37.3%	**30–45 FPS (CPU)** 120+ FPS (GPU)	**Ideal** — Zero latency, ultra-lightweight (3.2M params).
YOLOv8 / v11 Small	Single-Stage (Anchor-Free)	44.9% – 46.8%	20–30 FPS (CPU) 95+ FPS (GPU)	**High Accuracy** — Robust small object detection.
Faster R-CNN (ResNet-50)	Two-Stage (RPN + RoI)	40.2%	4–8 FPS (CPU) 18–25 FPS (GPU)	**Unsuitable** — Excessive lag; frames drop in live stream.
SSD MobileNet v2	Single-Stage (Anchor-Based)	22.1% – 31.2%	30–40 FPS (CPU) 70+ FPS (GPU)	**Suboptimal** — Fails to detect partially occluded phones.
RT-DETR (Baidu / Ultralytics)	Vision Transformer (DETR)	53.1% – 54.8%	3–6 FPS (CPU) 35–50 FPS (GPU)	**GPU Only** — High precision, but requires heavy VRAM.

2. Technical Advantages of YOLO in AI Proctoring

1. **Single-Stage Regression (Zero Latency):** Unlike two-stage detectors that generate regional proposals before classification, YOLO formulates object detection as a unified spatial regression problem. Bounding coordinates and confidence scores are generated simultaneously in a single forward pass (<10ms).
2. **Anchor-Free Head Architecture:** Older architectures relied on predefined anchor box ratios. YOLOv8 uses an anchor-free task-aligned assigner that predicts the center and dimensions directly. This significantly enhances detection of rotated smartphones, notebooks, and shifting candidate postures.
3. **CPU and Edge Deployment:** The lightweight backbone (C2f feature fusion blocks) enables smooth 30 FPS inference on standard laptops without requiring a dedicated CUDA GPU.
4. **Multi-Task Ecosystem:** Ultralytics YOLO provides a unified API supporting Object Detection, Pose Estimation, Hand Keypoint Tracking, and Instance Segmentation under a consistent interface.

3. Implementing Future Models (e.g., YOLOv26 or Custom Weights)

Model Lineage Clarification: The current official Ultralytics releases span from **YOLOv8** → **YOLOv9** → **YOLOv10** → **YOLOv11**. While there is no official 'YOLOv26' release yet in the academic community, the **EviGuard** codebase is engineered with a decoupled **Factory Pattern Architecture** that guarantees 100% plug-and-play compatibility with any future release.

How to Integrate a Future Model / Custom Checkpoint in EviGuard:
Step 1: *Place your new weights file (e.g., yolov26n.pt or custom_exam_proctor.pt) into the project root directory.*
Step 2: *Open config.yaml and update the model configuration section:*

```
detection:
  model_type: 'yolov8'  # Uses standard Ultralytics engine
  model_path: 'yolov26n.pt'  # Points to your new model
```

Step 3: *Restart the dashboard. DetectorFactory and EviGuardPipeline will automatically load the model weights, initialize inference tensors, and bind detections directly to the live HUD and Risk Scoring Engine without requiring any code refactoring.*

4. Dataset & Pretraining Analysis for YOLOv8

YOLOv8 base pretrained models (yolov8n.pt, yolov8s.pt, yolov8m.pt, yolov8l.pt, yolov8x.pt) were trained on extensive multi-million image benchmarks:

Dataset Benchmark	Image Count	Annotations / Classes	Role in Model Pipeline
MS COCO 2017 (Microsoft Common Objects in Context)	**330,000+ total images** • 118,287 Train • 5,000 Validation • 40,670 Test	**1.5 Million+** bounding box instances across **80 classes**. *(Includes Person, Phone, Book, Laptop)*	**Primary Object Detection Training:** Enables real-time classification and localization of candidates and unauthorized electronic items.
ImageNet-1k (Visual Recognition Challenge)	**1,280,000+** high-resolution training images	**1,000 categories** covering diverse real-world textures and objects.	**Backbone Feature Pretraining:** Initializes convolutional weights with rich multi-scale visual representations prior to COCO fine-tuning.

5. Summary & Academic Justification for EviGuard

1. Proven Balance: YOLOv8 provides the highest accuracy-to-compute ratio among modern detectors, making it the industry standard for real-time edge monitoring.

2. Extensibility: Because the system employs an abstract BaseDetector interface, future architectures (such as YOLOv10, YOLOv11, or a future YOLOv26) can be loaded with zero modification to business logic.

3. Zero-Lag Multi-Modal Integration: YOLO detections seamlessly fuse with MediaPipe 3D Head Pose, Gaze Estimator, and Dynamic Risk Scoring Engine for immediate incident flagging.